Plan

- Inner Product
- Gram-Schmidt Process

Inner Product spaces

Definition

Let V be a vector space over $\mathbb{F}$ ($\mathbb{F} = \mathbb{R}$ or$\mathbb{C}$). An inner product on V is a function $\langle\ ,\ \rangle : V \times V \to \mathbb{F}$ satisfying: for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V,\ c \in \mathbb{F}$

- $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, and $\langle \mathbf{v}, \mathbf{v} \rangle = 0 \iff \mathbf{v} = \mathbf{0}$.
- $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$.
- $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$.
- $\langle \mathbf{u}, c\mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$. [Note: $\langle c\mathbf{u}, \mathbf{v} \rangle = \overline{c}\langle \mathbf{u}, \mathbf{v} \rangle$.]

An inner product space is a vector space with an inner product.

The standard inner product on $\mathbb{R}^n$ (dot product on $\mathbb{R}^n$):

For $\mathbf{u} = [u_1, u_2, \ldots, u_n]^t, \mathbf{v} = [v_1, v_2, \ldots, v_n]^t \in \mathbb{R}^n$

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}\cdot\mathbf{v} := u_1 v_1 + u_2 v_2 + \ldots + u_n v_n = \mathbf{u}^t\mathbf{v}.$$

$\mathbb{R}^n$ with this inner product is called a Euclidean Space.

Definition (Standard inner product (dot product on $\mathbb{C}^n$))

For $\mathbf{u} = [u_1, u_2, \ldots, u_n]^t, \mathbf{v} = [v_1, v_2, \ldots, v_n]^t \in \mathbb{R}^n$

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} := \overline{u_1} v_1 + \overline{u_2} v_2 + \ldots + \overline{u_n} v_n = \mathbf{u}^* \mathbf{v}.$$

$\mathbb{C}^n$ with this inner product is called a Unitary Space.

We will restrict ourselves to $\mathbb{R}^n$ and $\mathbb{C}^n$ with dot products.

Definition

Let $\mathbf{u} = [u_1, u_2, \ldots, u_n]^t, \mathbf{v} = [v_1, v_2, \ldots, v_n]^t \in \mathbb{F}^n$ ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$).

The length or norm $\|\mathbf{u}\|$ of $\mathbf{u}$ is defined by

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{|u_1|^2 + |u_2|^2 + \ldots + |u_n|^2}.$$

Observe that $\|\alpha \mathbf{u}\| = |\alpha| \, \|\mathbf{u}\|$.

Definition

Vectors $\mathbf{u}$ and $\mathbf{v}$ are said to be orthogonal if $\mathbf{u}\cdot\mathbf{v} = 0$.

Example

- $[2, 1, -1]^t$ *and* $[0, 1, 1]^t$ *are orthogonal in* $\mathbb{R}^3$.
- $[1, i]^t$ *and* $[-1, i]^t$ *are orthogonal in* $\mathbb{C}^2$.

Result

For $\mathbf{u}, \mathbf{v} \in \mathbb{F}^n$

- $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ *(Triangular Inequality)*
- *if* $\mathbf{u}$ *and* $\mathbf{v}$ *are orthogonal, then*

  $$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \quad \textit{(Pythagorous Theorem)}$$

- $|\mathbf{u}\cdot\mathbf{v}| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$ *(Cauchy-Schwarz Inequality)*

  Equality occurs iff $\mathbf{u}$ *and* $\mathbf{v}$ *are linearly dependent, i.e., one of them is a multiple of the other.*

Proof. Exercise.

Definition

The angle θ between $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is defined by

$$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|}, \quad \theta \in [0, \pi] \qquad \text{(use Cauchy-Schwarz)}$$

Definition

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\} \subseteq \mathbb{F}^n$. Then S is said to be orthogonal if $\mathbf{v}_i\cdot\mathbf{v}_j = 0$ for all $i \neq j$.

An orthogonal basis is a basis which is an othrogonal set.

Example

- $\{[2, 1, -1]^t, [0, 1, 1]^t, [1, -1, 1]^t\}$ *is an orthogonal basis of* $\mathbb{R}^3$.
- $\{[1, i]^t, [-1, i]^t\}$ *is an orthogonal basis of* $\mathbb{C}^2$.
- $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ *is an orthogonal set in* $\mathbb{F}^n$ *(a very special one).*

Result

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is an orthogonal set of nonzero vectors, then S is linearly independent.

Proof.

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0} \implies \mathbf{v}_i \cdot (c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k) = 0$$
$$\implies c_i\|\mathbf{v}_i\|^2 = 0 \implies c_i = 0, 1 \le i \le k.$$

Definition

- A vector $\mathbf{u}$ is called an unit vector if $\|\mathbf{u}\| = 1$.
- An orthonormal set is an orthogonal set of unit vectors.
- An orthonormal set that forms a basis is called an orthonormal basis.

Example

The standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is an orthonormal basis of $\mathbb{F}^n$.

Example

- $\left\{ \begin{bmatrix} 1 \\ i \end{bmatrix}, \begin{bmatrix} -1 \\ i \end{bmatrix} \right\}$ *is an orthogonal basis in* $\mathbb{C}^2$.
- $\left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{-1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix} \right\}$ *is an orthonormal basis in* $\mathbb{C}^2$.

Exercise

Use the orthogonal basis $\{[2, 1, -1]^t, [0, 1, 1]^t, [1, -1, 1]^t\}$ *of* $\mathbb{R}^3$ *to find an orthonormal basis of* $\mathbb{R}^3$.

Result

Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ *be an orthogonal basis for a subspace* W *and let* $\mathbf{w} \in W$. *Then*

$$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k, \text{ where } c_i = \frac{\mathbf{v}_i \cdot \mathbf{w}}{\|\mathbf{v}_i\|^2}.$$

If B is orthonormal, then

$$\mathbf{w} = (\mathbf{v}_1 \cdot \mathbf{w})\mathbf{v}_1 + (\mathbf{v}_2 \cdot \mathbf{w})\mathbf{v}_2 + \cdots + (\mathbf{v}_k \cdot \mathbf{w})\mathbf{v}_k.$$

Orthogonal Complement of a set

Definition

Let S be a subset of $\mathbb{F}^3$.

- A vector **v** is said to be orthogonal to S if **v** is orthogonal to every vector in S.
- Orthogonal complement of S, denoted $S^{\perp}$, is defined as

$$S^{\perp} = \{\mathbf{v} \in \mathbb{F}^n : \mathbf{v} \cdot \mathbf{w} = 0 \text{ for all } \mathbf{w} \in S\}.$$

Example

- *The vector* $\mathbf{0}$ *is orthogonal to* S, *for any subset* S.
- *In* $\mathbb{R}^3$, *take* $S = \{\mathbf{e}_1\}$. *Then* $S^{\perp} = yz-$ *plane*.
- *In* $\mathbb{R}^3$, *take* $S = \{[1, 1, 1]^t\}$. *Then*
$$S^{\perp} = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : [1\ 1\ 1] \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \right\} = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x + y + z = 0 \right\}.$$

- In $\mathbb{C}^3$, take $S = \{[1,1,i]^t, [1,2i,3]^t\}$. Then
$$S^\perp = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : \begin{bmatrix} 1 & 1 & -i \\ 1 & -2i & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \right\}.$$

- In $\mathbb{C}^3$, take $W = \mathrm{span}\{[1,1,i]^t, [1,2i,3]^t\} = \mathrm{span}(S)$.
$$\text{Is } W^\perp = S^\perp = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : \begin{bmatrix} 1 & 1 & -i \\ 1 & -2i & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \right\}?$$

Result

For any subset S of $\mathbb{F}^n$

1. $S^\perp$ *is a subspace of* $\mathbb{F}^n$.
2. $S \cap S^\perp \subseteq \{\mathbf{0}\}$. [$\mathbf{v} \cdot \mathbf{v} = 0 \implies \mathbf{v} = \mathbf{0}$.]
3. $(span(S))^\perp = S^\perp$.

Result

Let W be a subspace of $\mathbb{F}^n$. Then

1. *If $B = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k\}$ is a basis for W, then*

   $\mathbf{v} \in W^\perp$ *if and only if* $\mathbf{v} \cdot \mathbf{w}_i = 0$ *for all* $i = 1, 2, \ldots, k$.

2. *If $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ and $\{\mathbf{v}_1, \ldots, \mathbf{v}_s\}$ are linearly independent sets in W and $W^\perp$, respectively, then*

   $\{\mathbf{u}_1, \ldots, \mathbf{u}_r, \mathbf{v}_1, \ldots, \mathbf{v}_s\}$ *is also linearly independent.*

Proof. (2) Since $W \cap W^\perp = \{\mathbf{0}\}$,

$$\begin{aligned}
& c_1\mathbf{u}_1 + \cdots + c_r\mathbf{u}_r + d_1\mathbf{v}_+ \cdots + d_s\mathbf{v}_s = \mathbf{0} \\
\implies \quad & c_1\mathbf{u}_1 + \cdots + c_r\mathbf{u}_r = -(d_1\mathbf{v}_+ \cdots + d_s\mathbf{v}_s) \in W \cap W^\perp = \{\mathbf{0}\} \\
\implies \quad & c_1 = \cdots = c_r = d_1 = \cdots = d_s = 0. \qquad \square
\end{aligned}$$

Q. Is it true that $W + W^\perp = \mathbb{F}^n$? If so, then $\mathbb{F}^n = W \oplus W^\perp$.

Orthogonal Projections

Definition

Let W be a subspace of $\mathbb{F}^n$ and let $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k\}$ be an orthogonal basis for W. If $\mathbf{v} \in \mathbb{F}^n$, then the orthogonal projection of $\mathbf{v}$ onto W is defined as

$$\mathrm{proj}_W(\mathbf{v}) = \frac{\mathbf{w}_1 \cdot \mathbf{v}}{\|\mathbf{w}_1\|^2}\mathbf{w}_1 + \frac{\mathbf{w}_2 \cdot \mathbf{v}}{\|\mathbf{w}_2\|^2}\mathbf{w}_2 + \cdots + \frac{\mathbf{w}_k \cdot \mathbf{v}}{\|\mathbf{w}_k\|^2}\mathbf{w}_k.$$

The component of $\mathbf{v}$ orthogonal to W is the vector

$$\mathrm{perp}_W(\mathbf{v}) = \mathbf{v} - \mathrm{proj}_W(\mathbf{v}).$$

Result

- $\mathbf{v} = proj_W(\mathbf{v}) + perp_W(\mathbf{v})$,
- $proj_W(\mathbf{v}) \in W, \;\; perp_W(\mathbf{v}) \in W^\perp$,
- $(proj_W(\mathbf{v})) \cdot (perp_W(\mathbf{v})) = 0$.
- $\mathbb{F}^n = W \oplus W^\perp$, *and so* $\dim(W) + \dim(W^\perp) = n$.

The fact $\mathbb{F}^n = W \oplus W^{\perp}$ is known as the Orthogonal Decomposition Theorem.

Result

Let W be a subspace of $\mathbb{F}^n$.

- *If $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k\}$ is an orthonormal basis for W, then*

$$proj_W(\mathbf{v}) \quad = \quad (\mathbf{w}_1.\mathbf{v})\mathbf{w}_1 + \ldots + (\mathbf{w}_k.\mathbf{v})\mathbf{w}_k.$$

- *If $\{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ is an orthonormal basis for $\mathbb{F}^n$, and $W = span\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k\}$, then $\{\mathbf{w}_{k+1}, \ldots, \mathbf{w}_n\}$ is an orthonormal basis for $W^{\perp}$.*
- $(W^{\perp})^{\perp} = W$.

Example

For $W = \{[x,y,z]^t \in \mathbb{R}^3 : x+y+z=0\}$,
$\{[-1,2,-1]^t, [1,0,-1]^t\}$ *is an orthogonal basis. We have*

$$\begin{aligned} proj_W([1,2,3]^t) &= [-1,0,1]^t, \text{ and} \\ perp_W([1,2,3]^t) &= [1,2,3]^t - [-1,0,1]^t = [2,2,2]^t. \end{aligned}$$

Exercise (Take home)

Let A *be an* $m \times n$ *complex matrix. Then show that*
(i) $(col(A))^\perp = null(A^*)$, *(ii)* $(row(\overline{A}))^\perp = null(A)$,
(iii) $row(\overline{A}) = null(A)^\perp$.
Deduce that $rank(A) + nullity(A) = n$.

Exercise (Take home)

Let U *be a subspace of* $\mathbb{F}^n$. *Suppose* $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ *forms a basis for* $U^\perp$. *Consider the* $k \times n$ *matrix* A *whose* i*-th row is* $\mathbf{v}_i^*$. *Show that* $U = null(A)$. *Conclude that every subspace of* $\mathbb{F}^n$ *is the solution space of a system of homogeneous linear equations.*

Exercise (Take home)

Let A and B be two $m \times n$ matrices and let the consistent linear systems $A\mathbf{x} = \mathbf{c}$ and $B\mathbf{x} = \mathbf{d}$ have the same solution set. Show that the matrix A is row equivalent to B.

Exercise (Take home)

Find a basis for $W^{\perp}$, where $W = \text{span}\left\{ \begin{bmatrix} 1 \\ -3 \\ 5 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 2 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 4 \\ -1 \\ 5 \end{bmatrix} \right\}$.

Result (**The Gram-Schmidt Process**)

Let $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}$ *be an ordered basis for a subspace* W *of* $\mathbb{F}^n$*. Define*

$$\mathbf{v}_1 = \mathbf{x}_1, \qquad W_1 = span(\mathbf{x}_1);$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \left(\frac{\mathbf{v}_1.\mathbf{x}_2}{\mathbf{v}_1.\mathbf{v}_1}\right)\mathbf{v}_1, \qquad W_2 = span(\mathbf{x}_1, \mathbf{x}_2);$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \left(\frac{\mathbf{v}_1.\mathbf{x}_3}{\mathbf{v}_1.\mathbf{v}_1}\right)\mathbf{v}_1 - \left(\frac{\mathbf{v}_2.\mathbf{x}_3}{\mathbf{v}_2.\mathbf{v}_2}\right)\mathbf{v}_2, \qquad W_3 = span(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3);$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

$$\mathbf{v}_k = \mathbf{x}_k - \left(\frac{\mathbf{v}_1.\mathbf{x}_k}{\mathbf{v}_1.\mathbf{v}_1}\right)\mathbf{v}_1 - \left(\frac{\mathbf{v}_2.\mathbf{x}_k}{\mathbf{v}_2.\mathbf{v}_2}\right)\mathbf{v}_2 - \ldots - \left(\frac{\mathbf{v}_{k-1}.\mathbf{x}_k}{\mathbf{v}_{k-1}.\mathbf{v}_{k-1}}\right)\mathbf{v}_{k-1},$$

$$W_k = span(\mathbf{x}_1, \ldots, \mathbf{x}_k).$$

Then $\{\mathbf{v}_1, \ldots, \mathbf{v}_i\}$ *is an orthogonal basis for* W_i*, for* $1 \leq i \leq k$*.*

In particular, $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ *is an orthogonal basis for* W*.*

Example

Apply the Gram-Schmidt process to find an orthonormal basis of the subspace spanned by $\mathbf{x}_1 = [1, -1, 1]^t$, $\mathbf{x}_2 = [0, 3, -3]^t$ *and* $\mathbf{x}_3 = [3, 2, 2]^t$.

- $\mathbf{v}_1 = \mathbf{x}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$,
- $\mathbf{v}_2 = \mathbf{x}_2 - \left(\frac{\mathbf{v}_1.\mathbf{x}_2}{\mathbf{v}_1.\mathbf{v}_1}\right)\mathbf{v}_1 = \mathbf{x}_2 - \left(\frac{-6}{3}\right)\mathbf{v}_1 = \begin{bmatrix} 0 \\ 3 \\ -3 \end{bmatrix} + \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$,
- $\mathbf{v}_3 = \mathbf{x}_3 - \left(\frac{\mathbf{v}_1.\mathbf{x}_3}{\mathbf{v}_1.\mathbf{v}_1}\right)\mathbf{v}_1 - \left(\frac{\mathbf{v}_2.\mathbf{x}_3}{\mathbf{v}_2.\mathbf{v}_2}\right)\mathbf{v}_2$

  $$= \mathbf{x}_3 - \left(\frac{3}{3}\right)\mathbf{v}_1 - \left(\frac{6}{6}\right)\mathbf{v}_2 = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}.$$
- $\frac{\mathbf{v}_1}{||\mathbf{v}_1||} = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$, $\frac{\mathbf{v}_2}{||\mathbf{v}_2||} = \begin{bmatrix} 2/\sqrt{6} \\ 1/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}$, $\frac{\mathbf{v}_3}{||\mathbf{v}_3||} = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$.
- An orthonormal basis is $\left\{ \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \begin{bmatrix} 2/\sqrt{6} \\ 1/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \right\}$.

- Given a set of vectors S, we can use Gram-Schmidt process to check its linear dependency.
- We can find an orthonormal basis B for span(S).
- The vectors in S corresponding to the elements of B are linearly independent.